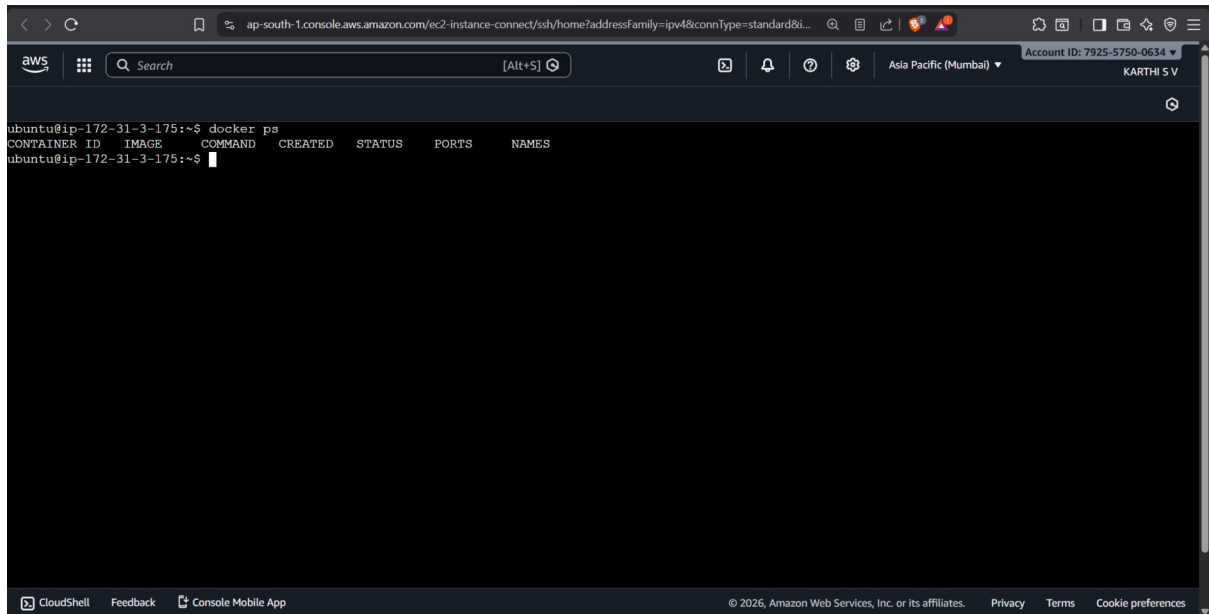


PROJECT - 3 REACT APPLICATION

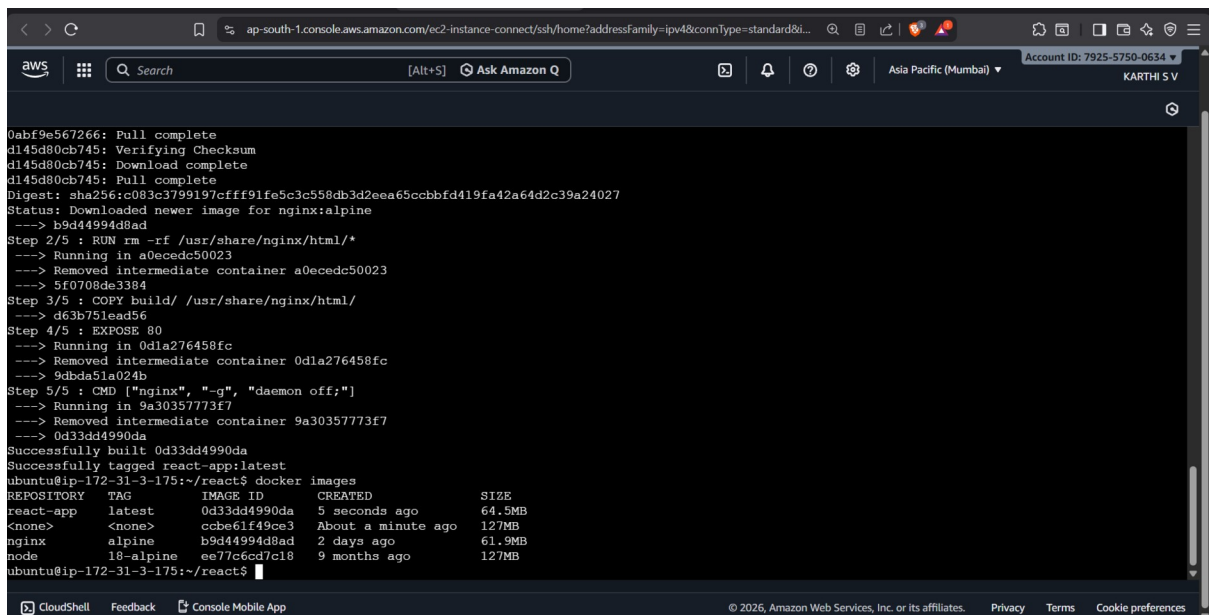
1 Docker setup



The screenshot shows the AWS CloudShell interface. The terminal prompt is `ubuntu@ip-172-31-3-175:~$`. The user has entered the command `docker ps`. The output shows a table with columns: CONTAINER ID, IMAGE, COMMAND, CREATED, STATUS, PORTS, and NAMES. The table is currently empty.

```
ubuntu@ip-172-31-3-175:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
ubuntu@ip-172-31-3-175:~$
```

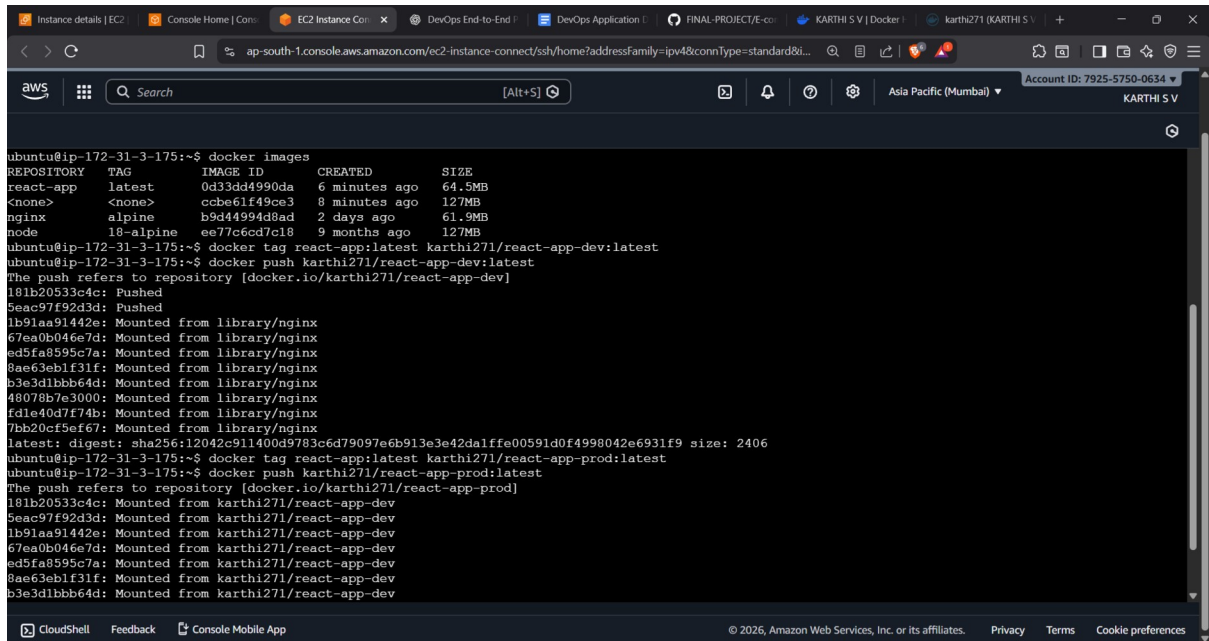
2 Docker image



The screenshot shows the AWS CloudShell interface. The terminal prompt is `ubuntu@ip-172-31-3-175:~$`. The user has entered the command `docker pull nginx:alpine`. The output shows the pull process for the `nginx:alpine` image. The terminal prompt is `ubuntu@ip-172-31-3-175:~$`. The user has entered the command `docker build -t react-app .`. The output shows the build process for the `react-app` image. The terminal prompt is `ubuntu@ip-172-31-3-175:~$`. The user has entered the command `docker images`. The output shows a table with columns: REPOSITORY, TAG, IMAGE ID, CREATED, and SIZE.

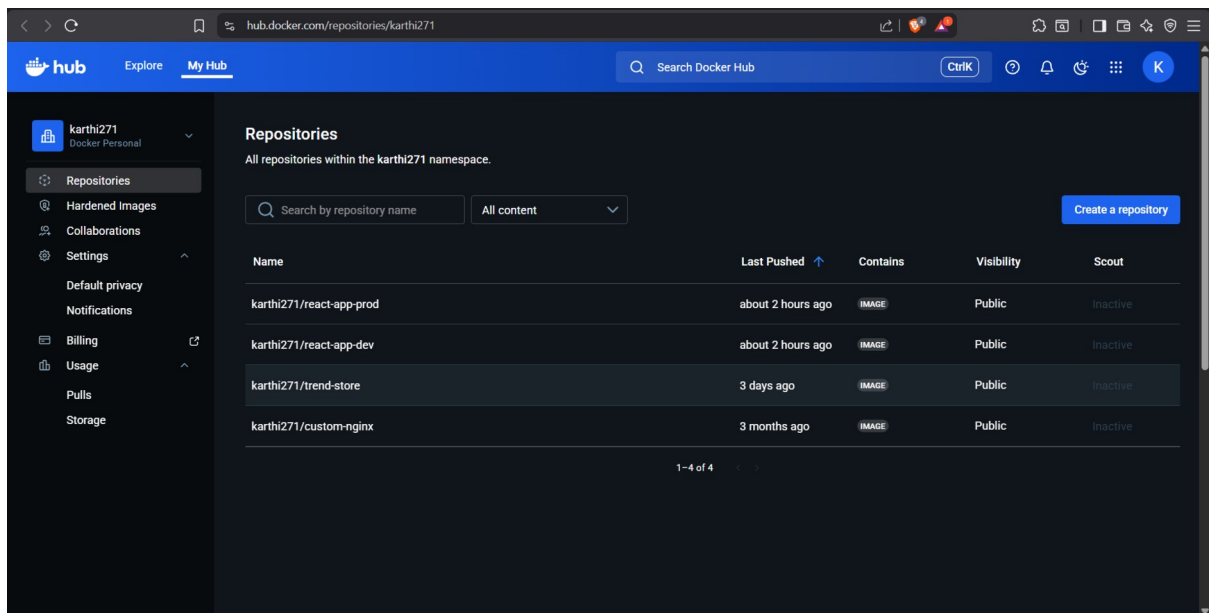
```
ubuntu@ip-172-31-3-175:~$ docker pull nginx:alpine
0abf9e567266: Pull complete
d145d80cb745: Verifying Checksum
d145d80cb745: Download complete
d145d80cb745: Pull complete
Digest: sha256:c083c3799197cfff91fe5c3c558db3d2eea65ccbbfd419fa42a64d2c39a24027
Status: Downloaded newer image for nginx:alpine
--> b9d44994d8ad
Step 2/5 : RUN rm -rf /usr/share/nginx/html/*
--> Running in a0ecedc50023
--> Removed intermediate container a0ecedc50023
--> 5f0708de3384
Step 3/5 : COPY build/ /usr/share/nginx/html/
--> d63b75lead56
Step 4/5 : EXPOSE 80
--> Running in 0d1a276458fc
--> Removed intermediate container 0d1a276458fc
--> 9dbda51a024b
Step 5/5 : CMD ["nginx", "-g", "daemon off;"]
--> Running in 9a30357773f7
--> Removed intermediate container 9a30357773f7
--> 0d33dd4990da
Successfully built 0d33dd4990da
Successfully tagged react-app:latest
ubuntu@ip-172-31-3-175:~$ docker images
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
react-app    latest    0d33dd4990da   5 seconds ago   64.5MB
<none>       <none>    ccb61f49ce3   About a minute ago   127MB
nginx        alpine    b9d44994d8ad   2 days ago     61.9MB
node         18-alpine ee77c6cd7c18   9 months ago   127MB
ubuntu@ip-172-31-3-175:~$
```

3 Docker hub push



```
ubuntu@ip-172-31-3-175:~$ docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
react-app            latest              0d33dd4990da       6 minutes ago      64.5MB
<none>               <none>             ccb61f49ce3        8 minutes ago      127MB
nginx                alpine             b9d44994d8ad       2 days ago         61.9MB
node                 18-alpine          ee77c6cd7c18       9 months ago       127MB
ubuntu@ip-172-31-3-175:~$ docker tag react-app:latest karthi271/react-app-dev:latest
ubuntu@ip-172-31-3-175:~$ docker push karthi271/react-app-dev:latest
The push refers to repository [docker.io/karthi271/react-app-dev]
181b20533c4c: Pushed
5eac97f92d3d: Pushed
1b91aa91442e: Mounted from library/nginx
67ea0b046e7d: Mounted from library/nginx
ed5fa8595c7a: Mounted from library/nginx
8ae63eb1f31f: Mounted from library/nginx
b3e3d1bbb64d: Mounted from library/nginx
48078b7e3000: Mounted from library/nginx
fd1e40d7f74b: Mounted from library/nginx
7bb20cf5ef67: Mounted from library/nginx
latest: digest: sha256:12042c911400d9783c6d79097e6b913e3e42dalffe00591d0f4998042e6931f9 size: 2406
ubuntu@ip-172-31-3-175:~$ docker tag react-app:latest karthi271/react-app-prod:latest
ubuntu@ip-172-31-3-175:~$ docker push karthi271/react-app-prod:latest
The push refers to repository [docker.io/karthi271/react-app-prod]
181b20533c4c: Mounted from karthi271/react-app-dev
5eac97f92d3d: Mounted from karthi271/react-app-dev
1b91aa91442e: Mounted from karthi271/react-app-dev
67ea0b046e7d: Mounted from karthi271/react-app-dev
ed5fa8595c7a: Mounted from karthi271/react-app-dev
8ae63eb1f31f: Mounted from karthi271/react-app-dev
b3e3d1bbb64d: Mounted from karthi271/react-app-dev
```

4 Docker prod and dev



5 Jenkins setup

```
ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressfamily=ipv4&connType=standard8d...
AWS Search [Alt+S]
Account ID: 7925-5750-0634 KARTHI S V

ubuntu@ip-172-31-3-175:~$ sudo nano /etc/default/jenkins
ubuntu@ip-172-31-3-175:~$ sudo systemctl reset-failed jenkins
ubuntu@ip-172-31-3-175:~$ sudo systemctl daemon-reexec
ubuntu@ip-172-31-3-175:~$ sudo systemctl daemon-reload
ubuntu@ip-172-31-3-175:~$ sudo systemctl restart jenkins
ubuntu@ip-172-31-3-175:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-01-12 11:36:36 UTC; 4s ago
     Main PID: 7168 (java)
       Tasks: 47 (limit: 2213)
      Memory: 633.7M (peak: 656.7M)
         CPU: 22.398s
    CGroup: /system.slice/jenkins.service
            └─7168 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jan 12 11:36:30 ip-172-31-3-175 jenkins[7168]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jan 12 11:36:30 ip-172-31-3-175 jenkins[7168]: [LF]>
Jan 12 11:36:30 ip-172-31-3-175 jenkins[7168]: [LF]> *****
Jan 12 11:36:30 ip-172-31-3-175 jenkins[7168]: [LF]> *****
Jan 12 11:36:30 ip-172-31-3-175 jenkins[7168]: [LF]> *****
Jan 12 11:36:36 ip-172-31-3-175 jenkins[7168]: 2026-01-12 11:36:36.646+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Comple
Jan 12 11:36:36 ip-172-31-3-175 jenkins[7168]: 2026-01-12 11:36:36.680+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is s
Jan 12 11:36:36 ip-172-31-3-175 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jan 12 11:36:38 ip-172-31-3-175 jenkins[7168]: 2026-01-12 11:36:38.348+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained
Jan 12 11:36:38 ip-172-31-3-175 jenkins[7168]: 2026-01-12 11:36:38.348+0000 [id=49] INFO hudson.util.Retrier#start: Performed the actions

ubuntu@ip-172-31-3-175:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
14c53bb83c954ff5ad772bc8a77cd910
ubuntu@ip-172-31-3-175:~$
```

6 Jenkins credentials

52.66.248.32:8080/manage/credentials/store/system/domain/

Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestr...

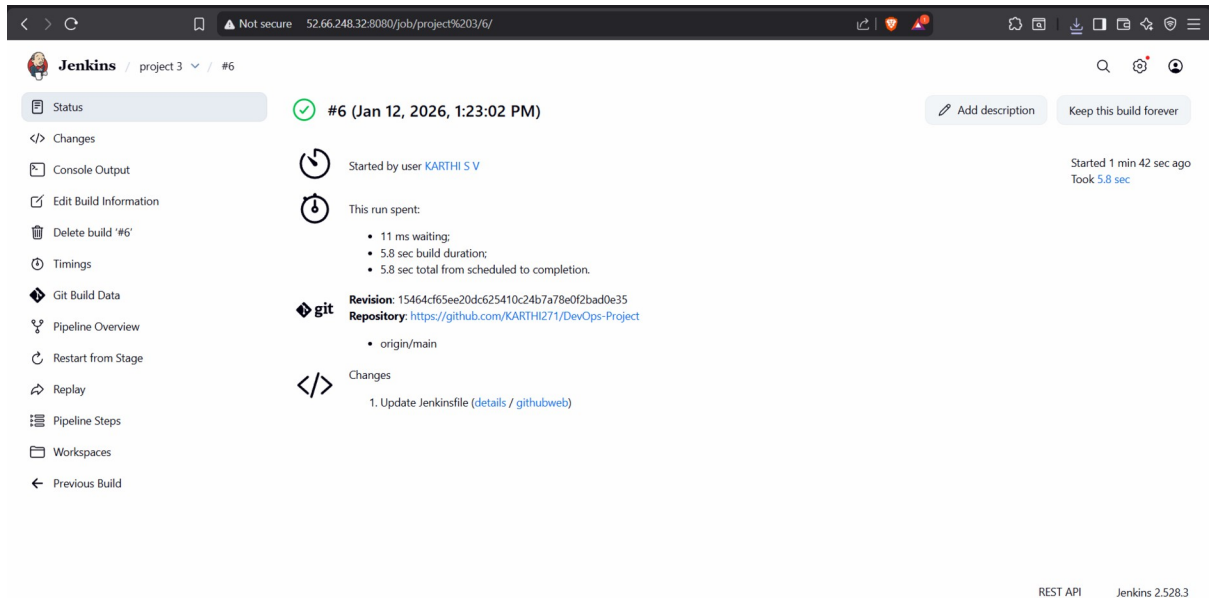
Global credentials (unrestricted) [+ Add Credentials](#)

Credentials that should be available irrespective of domain specification to requirements matching.

 karthi271/***** (docker login -u karthi271)
dockerhub-creds - docker login -u karthi271

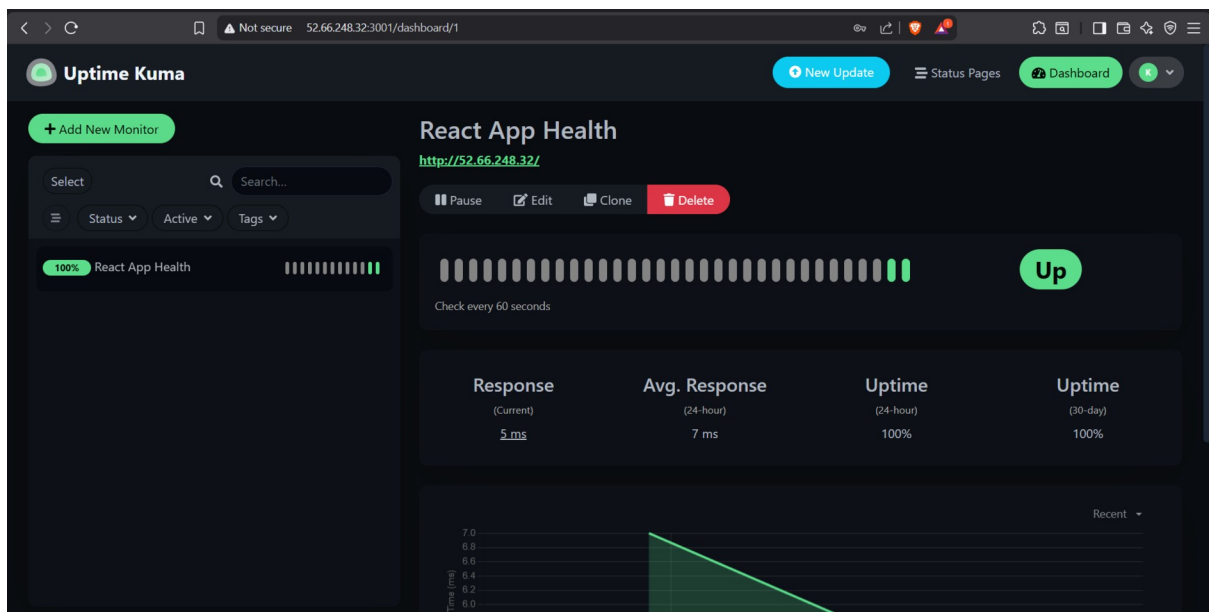
REST API Jenkins 2.528.3

7 Jenkins build success



The screenshot shows the Jenkins web interface for a successful build. The browser address bar indicates the URL is 52.66.248.32:8080/job/project%203/6/. The Jenkins logo and 'project 3' are visible in the top left. The build status is '#6 (Jan 12, 2026, 1:23:02 PM)' with a green checkmark. The left sidebar contains navigation links: Status, Changes, Console Output, Edit Build Information, Delete build '#6', Timings, Git Build Data, Pipeline Overview, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main content area shows build details: 'Started by user KARTHI S V', 'This run spent: 11 ms waiting, 5.8 sec build duration, 5.8 sec total from scheduled to completion.', 'Revision: 15464cf65ee20dc625410c24b7a78e0f2bad0e35', 'Repository: https://github.com/KARTHI271/DevOps-Project', and 'Changes: 1. Update Jenkinsfile (details / githubweb)'. The bottom right corner shows 'REST API' and 'Jenkins 2.528.3'.

8 Monitoring



The screenshot shows the Uptime Kuma monitoring dashboard. The browser address bar indicates the URL is 52.66.248.32:3001/dashboard/1. The Uptime Kuma logo is in the top left. The dashboard has a dark theme. On the left, there's a sidebar with a search bar and filters for Status, Active, and Tags. A monitor named 'React App Health' is listed with a 100% status. The main content area shows the 'React App Health' monitor details, including the URL 'http://52.66.248.32/'. The monitor is 'Up' and has a green status bar. Below the status bar, there are four metrics: Response (Current: 5 ms), Avg. Response (24-hour: 7 ms), Uptime (24-hour: 100%), and Uptime (30-day: 100%). At the bottom, there's a line graph showing the response time over time.

9 Online shop

